DEPARTMENT OF METEOROLOGY | UNIVERSITY OF READING

Hybrid Physics–ML Ensembles for Data Assimilation in Chaotic Systems

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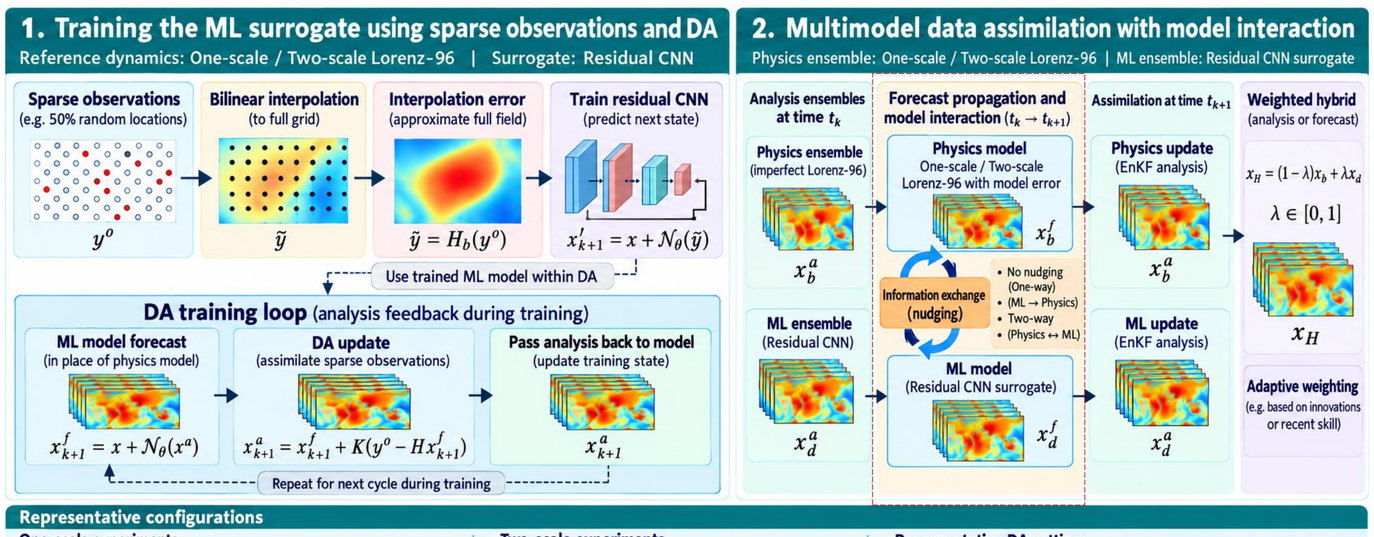
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INTRODUCTION

There is growing recognition in the weather and climate prediction community that combining multiple imperfect models offers the potential for more skilful forecasts, particularly when these models capture complementary aspects of the underlying physical system and exhibit different error characteristics.

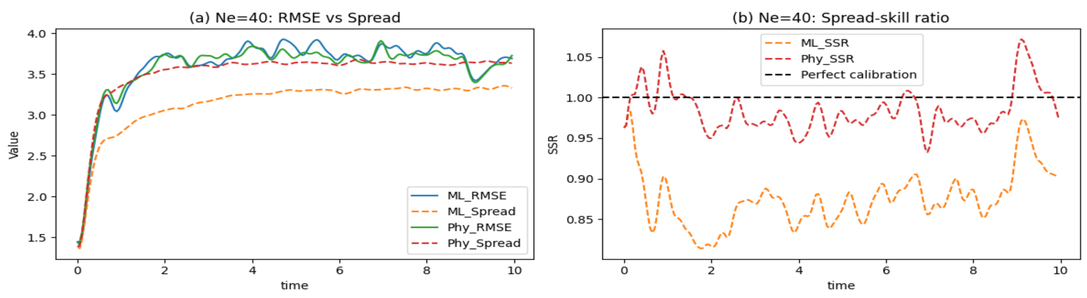
Recent advances in machine learning (ML) forecasting present new opportunities for multi-model prediction. While physics-based models represent atmospheric dynamics through well known governing equations, ML models learn these dynamics from data. Their distinct representations and error characteristics motivate their integration within ensemble data assimilation frameworks, although exploiting their complementary strengths remains under explored.

APPROACH



RESULTS & DISCUSSION

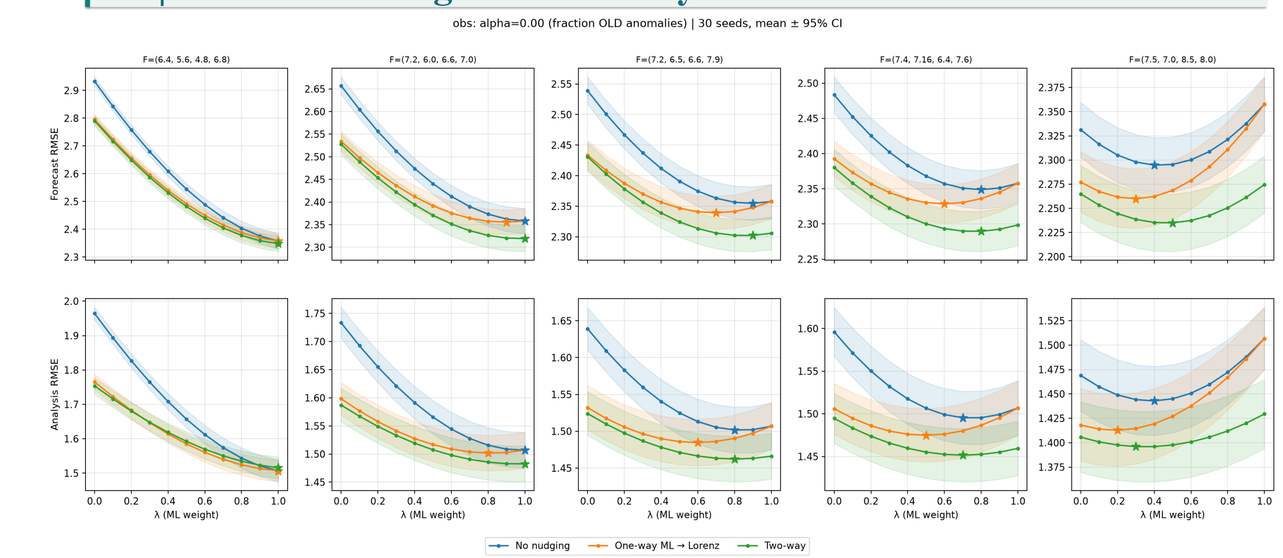
01 | Ensemble spread and RMSE



The physics-based and MSE-trained ML ensembles were initialised from identical perturbed states and integrated freely without data assimilation. The ML surrogate exhibits under-dispersion, with ensemble spread smaller than RMSE, reflecting the limitations of deterministic MSE training. During DA, we address model-error uncertainty through additive covariance inflation, using an innovation-estimated diagonal model-error covariance (Q).

$$d_k = y_k - H\bar{x}^f_k,\ S_k = \rho_S S_{k-1} + (1-\rho_S) d_k d_k^\top$$
$$HQH^\top \approx S_k - HP^f_k H^\top - R,\ P^f_{infl} = P^f + Q,$$

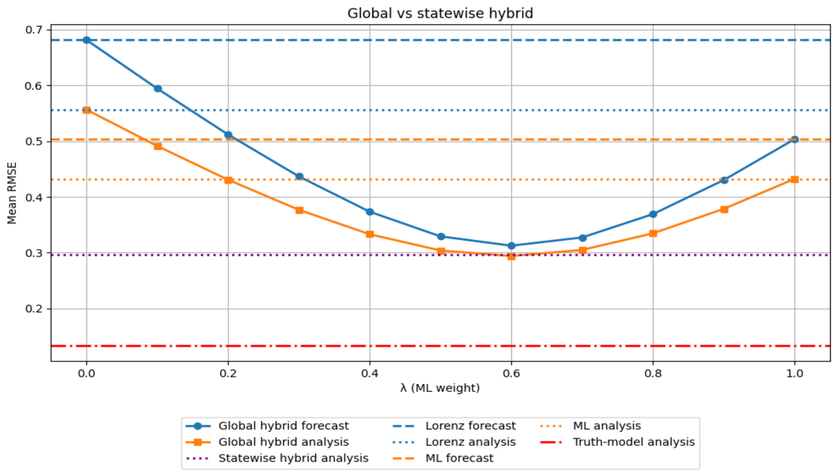
02 | One Scale Weighted Mean Hybrid



To exploit complementary model errors, we construct a weighted hybrid forecast by combining the physics-based and ML predictions. The optimal global weight is obtained by minimising the expected squared error, accounting for the individual model errors and their cross-correlation.

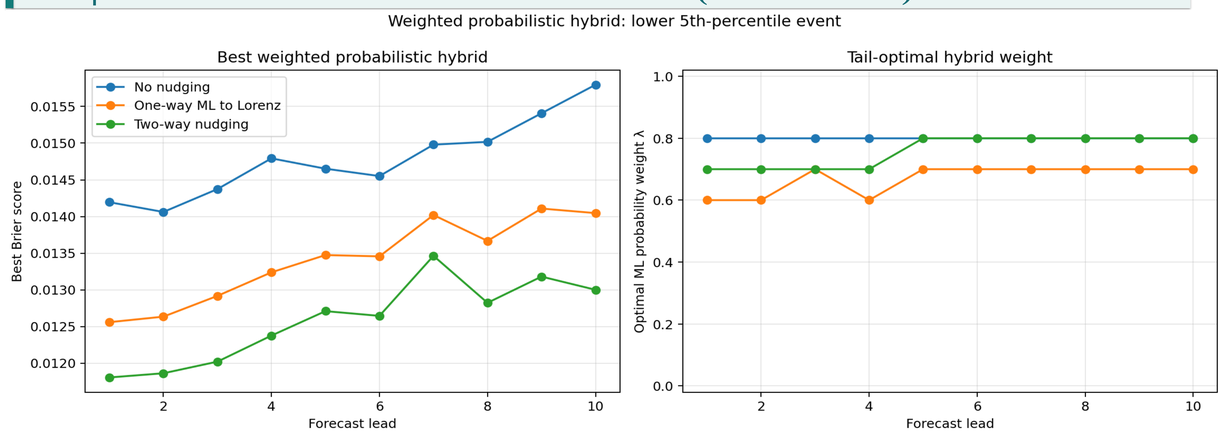
$$x_{hyb} = (1-\lambda)x_{lor} + \lambda x_{ml}$$
$$J(\lambda) = (1-\lambda)^2 A + \lambda^2 B + 2\lambda(1-\lambda)C$$
$$A = \mathbb{E}[\|e_{lor}\|^2], \quad B = \mathbb{E}[\|e_{ml}\|^2], \quad C = \mathbb{E}\left[e_{lor}^\top e_{ml}\right]$$
$$\lambda^* = \frac{A-C}{A+B-2C}$$

03 | Two Scale Weighted Mean Hybrid



The results show that a global ML weight of approximately ($\lambda = 0.6$) minimises both forecast and analysis RMSE. State-wise weighting provides negligible additional improvement, suggesting that a single global weight is sufficient for this experiment.

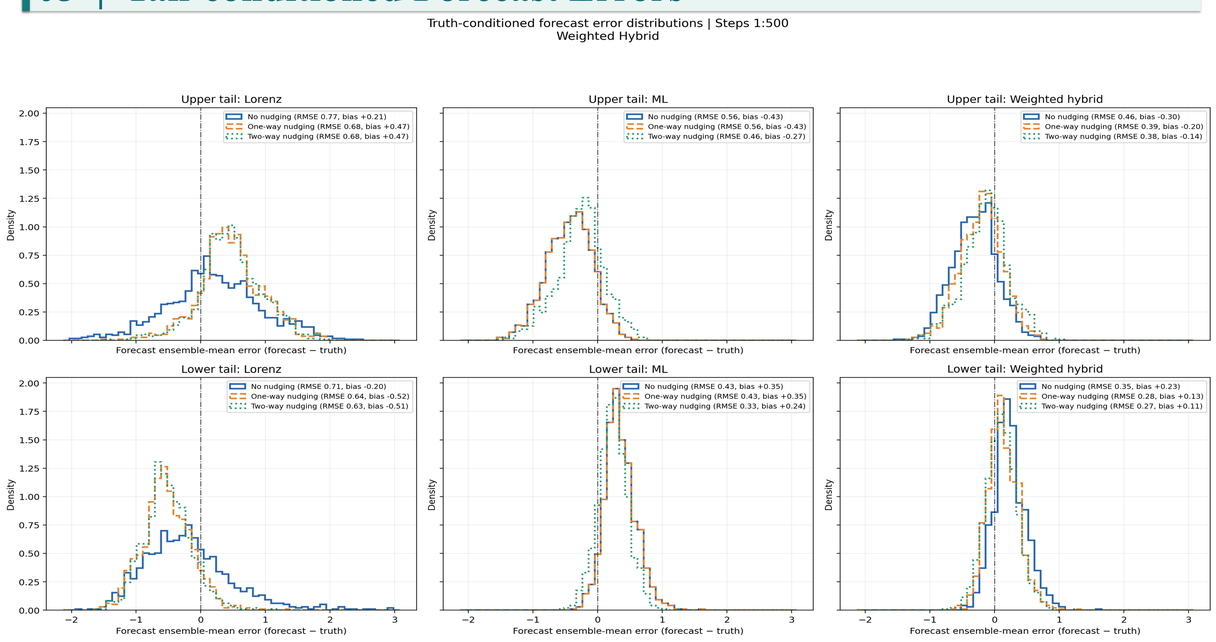
04 | Probabilistic Extreme-event Skill (Brier Score)



We evaluate probabilistic forecast skill for extreme events defined by the lower 5th percentile of the truth distribution using the Brier Score (BS). Two-way nudging consistently produces the lowest Brier Scores across forecast lead times, outperforming both one-way and no nudging. The optimal hybrid weights favour the ML model $\left((\lambda \approx 0.7\text{–}0.8)\right)$, highlighting its contribution to extreme-event prediction.

$$BS = \frac{1}{N}\sum_{i=1}^{N}(p_i - o_i)^2$$

05 | Tail-conditioned Forecast Errors



The physics-based and ML models exhibit distinct conditional biases in the upper and lower extremes, highlighting their complementary error characteristics. Two-way nudging reduces the weighted hybrid's forecast errors and brings its conditional biases closer to zero, despite persistent biases in the individual models. This demonstrates the benefit of cross-model information exchange for predicting extreme states.

06 | Key Findings And Future Work

- Combining physics-based and ML ensembles within the multi-model DA framework improves forecast and analysis accuracy relative to the individual models in our hybrid experiments. One-way and two-way nudging allow physics-based and ML ensembles to exchange information during forecast propagation. The benefit depends on both individual model errors and their cross-correlation. Strong information exchange risks synchronising the models, reducing error diversity and potentially limiting the gains from hybridisation.

- **Future Work: Extending multi-model DA to particle filters**
  Extend the multi-model framework to particle filters by combining physics-based and ML forecast ensembles within a unified particle representation. Investigate whether cross-model information exchange and ML-informed proposal distributions improve particle diversity, reduce weight degeneracy and enhance probabilistic forecast skill in strongly nonlinear and non-Gaussian systems.

References

Bach, E., & Ghil, M. (2023). A multi-model ensemble Kalman filter for data assimilation and forecasting. Journal of Advances in Modeling Earth Systems, 15(1), e2022MS003123.

Brajard, J., Carrassi, A., Bocquet, M., & Bertino, L. (2020). Combining data assimilation and machine learning to emulate a dynamical model from sparse and noisy observations: A case study with the Lorenz 96 model.

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